Project-III Mid-Term Progress Report on

Blood Axis

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Submitted By

Nishant Shah (021-351)

Shree Ram Sah (021-380)

Srijan Gajurel (021-383)

Supervised By: Tara Chandra Shrestha

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Abstract

Blood donation is a key part of health care, yet many areas face delays because donors, blood banks, and hospitals are not well connected. Paper based or partly digital systems make it hard to keep correct stock details, check blood group availability on time, and follow each donation from donor to patient. Blood Axis aims to solve these problems by creating a web based Blood Bank Management System that links donors, blood banks, hospitals, and an admin in one place. The system was built using modern web tools for the database, server and user interface to support secure login, role based access and live data updates. Donors can register, record donations, see past donation history, and check the next allowed donation date. Organization manage incoming and outgoing blood units with checks that stop errors in stock levels, while hospital can request blood by group and track usage from organization. An admin oversees approvals, users, and request to keep the system trusted and safe. The system also provides clear charts of blood stock and includes a local blood bank with location details to help users find nearby services. This will show better stock accuracy, faster response in urgent cases, and clearer coordination among all roles. The system improves transparency and supports safer and more efficient blood donation and transfer services

[keywords: Online Blood Donation System, Blood Inventory Tracking, Emergency Blood Availability, Web Based Healthcare System]

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Abbreviations

Abbr.	Full Name
API	Application Programming Interface
SQL	Structured Query Language
JSON	JavaScript Object Notation
JWT	Json Web Token

Chapter 1: Introduction

1.1 Overview

Blood donation is essential aspect of healthcare as it saves countless lives each year. In many countries blood transfusion are the only treatment option for patients who have lost a lot of blood due to accident, surgeries, or medical conditions such as anemia and cancer. However, many places in the world face a serious lack of blood supply making it difficult for healthcare providers and patients. According to the Nepal Red Cross Society(NRCS), Nepal require around 300,000 units of blood every year, but the actual collection often falls short by 10-20% [1]. In remote and rural areas, the shortage is even more critical, putting lives at risk due to delays in blood availability. In addition, the traditional methods of blood donation have several limitations, including the need for donors to physically visit blood bank or donation centers, which can be time consuming and inconvenient. To address these challenges, there is a growing need for an online blood donation system that connects blood donors with patients or hospitals in need of blood. Blood Axis provides a platform for donors to register, and for users to search and request for blood donations online. This system can simplify the process of blood donation, make it more readily available to those who need it, and potentially encourage more people to donate blood. Built using

1.2 Problem Statement

There are a number of challenges associated with the traditional methods of blood donation, which an online system can help to address. These include delays in finding compatible donors, time-consuming paperwork, and inefficient communication between donors and healthcare providers. In emergency situations, such delays can be life threatening, especially when there is an urgent need for specific blood type. Manual registration process and the lack of donor engagement also contribute to lower donation rates [2]. Many potential donors are discouraged by the inconvenience of visiting blood banks or hospitals, which often involves long waits and lengthy procedures. This results in gap between the blood required and the amount actually collected. Blood Axis can address these problems by providing a platform for donors to register, and for users to search and request for blood donations online. The system can simplify the process of blood donation, making it more accessible to those in need and potentially increase the number of blood donors. In summary the current challenges in traditional blood donation methods like delays, low donor participation, inefficiencies can be tackled by Blood Axis which can enhance blood donation services and potentially save more lives

1.3 Objectives

- To create a secure and user-friendly online platform that secures blood donation, improves donor-recipient matching, and raises awareness to support an efficient and responsive healthcare system

1.4 Aim

Blood Axis aims to develop a secure and efficient online blood donation system that bridges the gap between donors and patients in need. This project aims to eliminate the delays and inconveniences of traditional manual procedures by providing a faster, more accessible and reliable digital solution. It is driven by the goal of increasing donor participation, simplifying the process for both donors and hospitals, and ultimately saving lives through timely transfusions.

1.5 Motivation

Blood Axis is motivated by the urgent need to address the persistent shortage of blood supply, improve donor-recipient connectivity, and enhance the efficiency of the blood donation process.

1.6 Applications and Scope

Blood Axis has a wide range of applications and scope, directly impacting the blood donation ecosystem and healthcare infrastructure by connecting donors, recipients, and medical institutions. Some applications include:

- Emergency Blood Management

  Blood Axis can be used in hospitals, clinics, and emergency response centers that require immediate access to blood ensuring quicker and more efficient transfusion processes.

- Donor Patient connectivity

  Blood Axis connects registered blood donors with patients or hospitals in need, reducing delays and improving the accuracy of donor-recipient matching.

- Centralized Blood Bank Operations

  It enables blood banks to manage donor records, blood stock levels, and expiry tracking efficiently through a centralized digital system.

The scope of Blood Axis extends to improving the overall efficiency, accessibility and safety of blood donation and transfusion services. The key features includes:

- Developing a user-friendly online platform for donors, recipients, and blood bank administrators using React JS, NodeJS, MongoDB and Express JS.
- Providing a centralized dashboard for blood banks to manage donations, donor records, blood inventory and expiry tracking efficiently.
- Enhancing the communication through email notification for donation drives, urgent request, and donor reminders
- Integrating location-based search to help users find nearby blood banks and available donors quickly.

1.7 Feasibility Study

A feasibility study is performed to determine whether the purposed system is viable considering the technical, operational, and economic factors. After going through feasibility study, we can have a clear-cut view of the system's benefits and drawbacks

1.7.1 Economical Feasibility

The necessary hardware and software are available in the market at a low cost, the system only needs an initial setup cost. After that, there are no major extra cost for upgrades or improvements. This makes the system cost-effective. Overall, the system is practical and affordable, which supports the decision to go ahead with its design and development

1.7.2 Technical Feasibility

The proposed system is developed using ReactJS, Laravel, Bootstrap and MySQL. It requires a local or web server to process user requests, such as XAMPP or any other Personal Web Server. Users can access the system through a web browser, which is available on all major operating system like Windows, Linux, and MacOS. The development was done on Windows 11 machine, which is user-friendly and widely used. All the necessary hardware and software are readily available in the market. Hence, the system is technically feasible.

1.7.3 Operational Feasibility

Blood Axis is operationally feasible because the customer is benefited more than most of his time is saved. The customer is serviced at his place of work. The cost of the proposed system is almost negligible when compared to the benefits gained.

Chapter 2: Literature Review

Blood is a vital resource that can save lives during medical emergencies, surgeries, and treatments. In many situations, finding the right blood group at the right time can be difficult and time-consuming. Traditional blood bank systems often depend on manual records, phone calls, and in-person visits, which can delay the process of donating or receiving blood. With the rise of digital technologies, online blood bank management systems have been developed to make this process faster, easier and more organized. Blood Axis provides an online platform to connect donors, recipients, hospitals, and blood bank in real time. Users can register as donors, request blood, check blood stock availability and receive notifications. This improves communication, saves time and increases the chances of saving lives. Several existing system have been already developed. However, most existing systems face problems such as outdated user interfaces and lack of real time updates. Some platforms do not verify donor information properly which can lead to trust issues. Others fail to provide quick notifications, making it hard to respond during emergencies. Many systems also lack proper integration with hospitals or do not offer location based search features, making it harder for users to find nearby donors or blood banks. Several researchers and developers have made significant steps towards creating digital solutions for managing blood banks. These systems are designed to improve accessibility, accuracy and efficiency in blood donation and transfusion processes

2.1 Previous Works

Several research studies have explored the importance and development of Blood Bank Management System emphasizing the need to digitize and streamline blood donation and transfusion processes.

In their study titled "A Study on Blood Bank Management", Teena, C.A., Sankar, K., and Kannan, S. introduced a centralized information system designed for efficient donor and patient record management. Their proposed system incorporated secure login access, blood stock monitoring, donor registration, and recipient tracking. This adaptable system was targeted toward various blood banks and healthcare personnel, such as administrators, doctors, and receptionists. It aimed to reduce manual errors and provide better accessibility for managing blood donations and reservations. [3]

A similar approach was taken by Kumar, Singh, and Ragavi (2017) in their study titled *"Blood Bank Management System,"* where they developed a web-based platform to facilitate national-level blood distribution. By adapting technologies like ASP.NET and PHP, the platform allowed hospitals to access donor databases in real time, improving nationwide blood availability. [4]

Further, Liyana F. (2017), in her paper *"Blood Bank Management System Using Rule-Based Method,"* proposed a rule-based web application that could automate decision-making for blood requests and donor communication. It prioritized features like tracking blood bag expiration and notifying donors when a specific blood group was in short supply. Her incremental model ensured adaptability and user feedback integration. [5]

While the above mentioned research laid foundational frameworks, various modern platforms like Nepali Blood Donors(Nepal), e-Raktkosh(India), Blood Donor Finder(Red Cross), and myBlood attempt to bring these ideas into practice through nationwide blood donor databases and real time availability tracking. Some notable platforms include:

e-Raktkosh(India) is a government initiative that connects blood banks across the country with real time data. While it enables hospital integration and donor registration, the platforms lacks effective donor engagement tools like timely reminders. [6]

Red Cross Blood Donor App(USA) is a mobile app that allows users to find donation centers, schedule appointments and track donations. However, it is primarily focused on logistics and lacks personalized dashboards that could promote donations through engagement and transparency. [7]

Some other platforms like Friends2Support provide donor search features and basic contact tools. However, their system rely on manual inputs, static list and SMS based contact making them inefficient for critical situation. [8]

Some of the identified gaps include failure to provide real time emergency support by not alerting nearby donors instantly. Unlike other personal apps, most of the apps don't offer personalized dashboard for donors to track their history, health status, eligibility or receive recognition for their contribution. Moreover most of these platforms often lack community features like groups or forums limiting donor engagement and motivation for repeat donations. The analysis presents a need for smarter blood bank system that not only streamlines administrative tasks but also engages user through real time updates, personalized dashboards and motivational elements.

Chapter 3: System Design

Blood Axis is a web based blood management system with ai forecasting built using MongoDB, Express.js, React, Node.js and Python to manage blood donation, blood request, stock tracking among donors, blood banks, hospitals, and admins. The system uses one common user structure with role based dashboards for four roles admin, donor, organization and hospital. The admin manages user approvals, controls accounts, and monitors all blood requests. Organizations handle blood stock by recording incoming donations, supplying blood to hospitals, and viewing stock details by blood group. Donors can signup, log in, donate blood to chosen organizations, and view their donation history, total donated units, last donation date, and next allowed donation date. Hospitals place blood requests, follow request status, and view records of blood received. The front end is built with React and supporting tools to provide a clear and responsive interface based on user role. The back end uses Node.js and Express to provide secure APIs connected to MongoDB for storing users, stocks, and requests with safe login and password protection. The system also includes a demand forecast feature that uses past stock and request data to estimate short term blood needs, helping reduce shortages and improve planning for all the users.

3.1 Block diagram

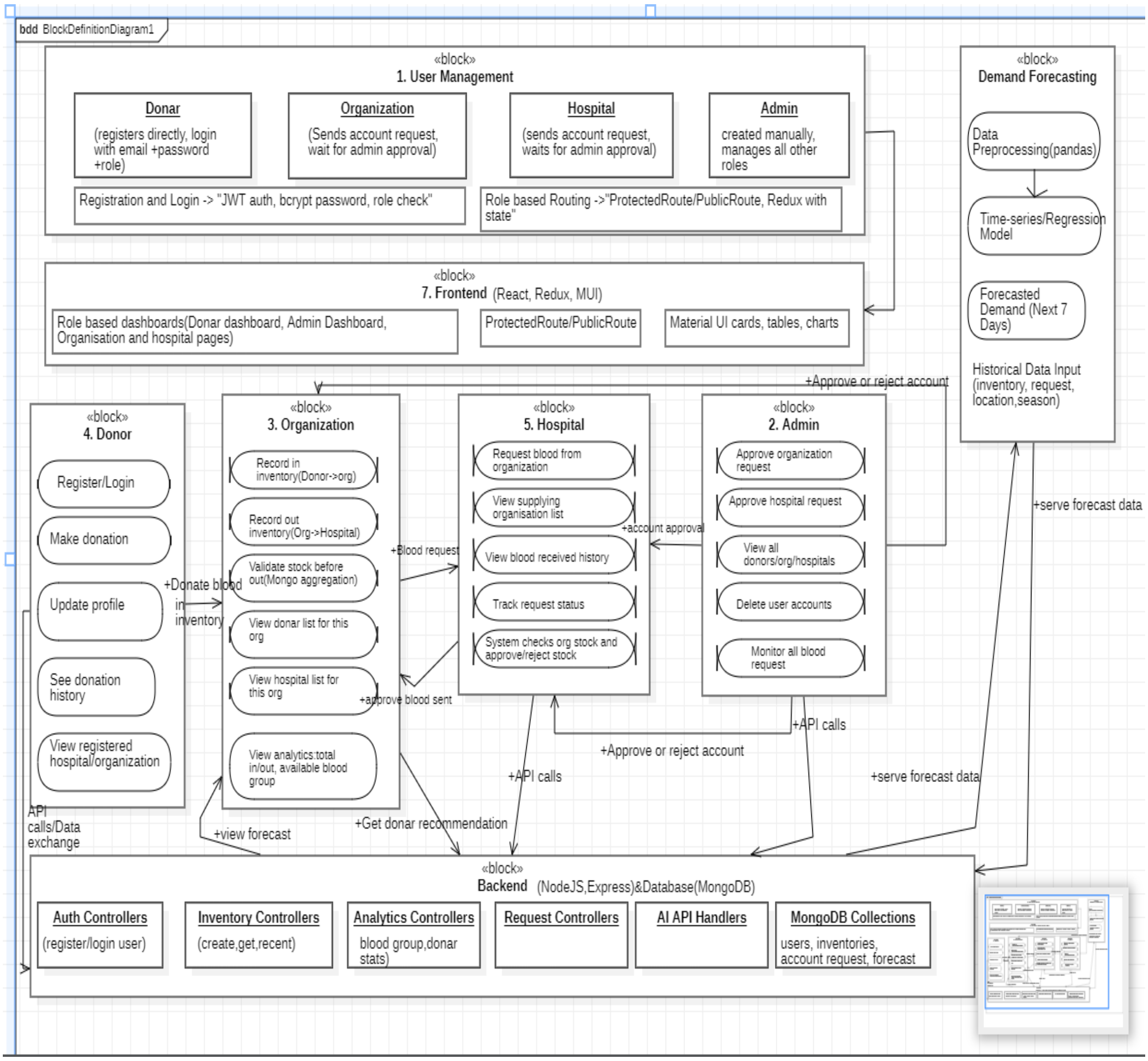

Figure 3 1:Block Diagram

This block diagram explains the full structure of Blood Axis with an added AI based demand forecast feature. The system supports four roles donor, organization, hospital and admin which are handled through a central user management layer that manages sign up role assignment and secure login using token based login password hashing and role checks. A React based frontend provides separate dashboard for each role protected pages and simple charts for data viewing and actions. The donor module lets users signup, login, donate blood, update profiles, and view donation history. The organization module handles blood stock by recording incoming donation history. The organization module handles blood stock by recording incoming donations and outgoing supply to hospitals checking available units showing linked donors and hospitals and displaying stock and usage data by blood group. The hospital module allows hospitals to request blood, track request status, see received history and view supplying organizations. The admin module controls approvals user management and system wide request tracking. All modules connect to a Node and Express backend that manages login inventory analytics and requests using a database for users stock requests and forecast records. A demand forecast unit uses past stock requests location and seasonal data to create seven day demand estimates to support better blood collection and supply planning

3.2 Use Case Diagram

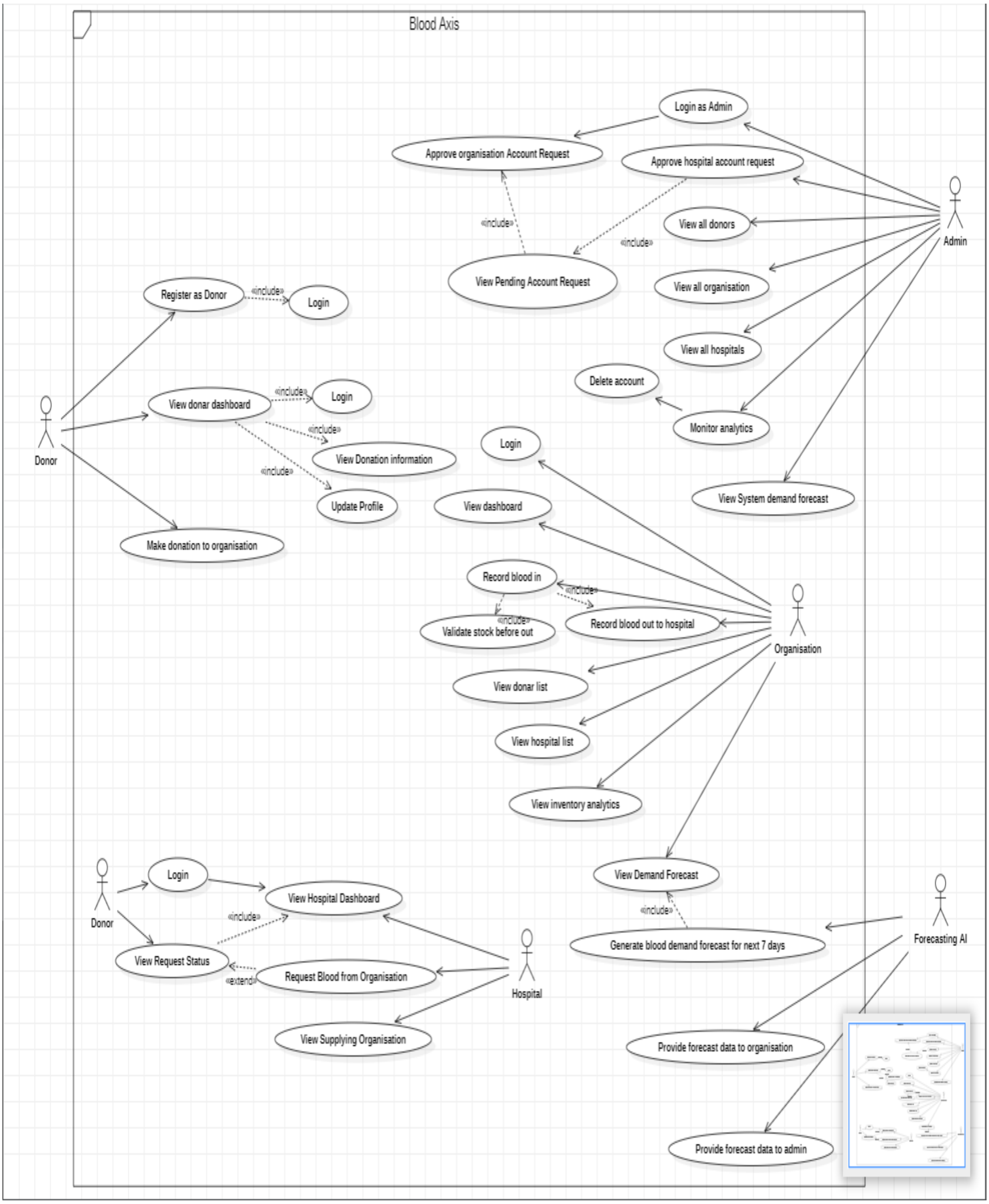

Figure 3 2:Use Case Diagram

A use case diagram depicts the possible interactions of various users of the system with the system itself. The use cases are represented by ellipses while the users (also called as actors) are depicted as stick figures. The use case diagram explains how the Blood Axis system works for four roles donor, organization, hospital, and admin along with a forecasting AI. Donors log in to access their dashboard where they can update profiles, view donation details, and donate blood to selected organizations. Organizations login to manage their dashboards where they record blood received from donors, supply blood to hospitals, check stocks before supply, and view donor ,hospital, and inventory data. Hospital log in to request blood track request status and see which organizations supply them. The admin logs in with full control to approve or reject organization and hospital accounts manage all users view system data and see demand forecast results. The forecasting AI provides a seven day blood demand estimate that helps organizations and the admin plan blood collection and stock levels better.

3.3 ER Diagram

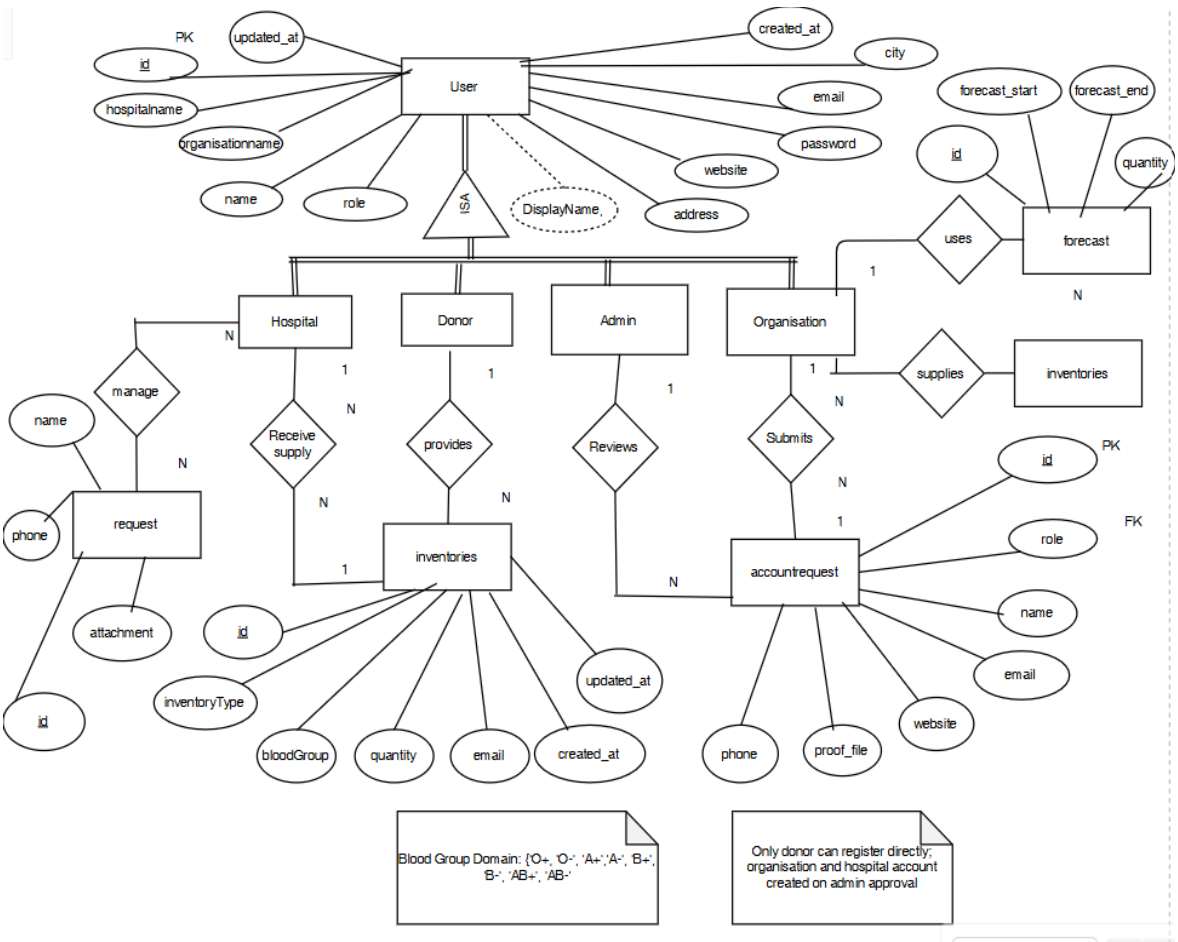

Figure 3 3:ER Diagram

Entity-Relationship (ER) explains how data is stored and linked in the Blood Axis system. A main User entity holds shared details for all roles and is divided into donor, hospital, organization and admin. Donor, hospital, organization data connect to inventory records which store all blood donations and supply details by blood group and amount. Hospitals send blood requests that are saved in the requests where admins approve or reject them to control valid system access. The forecast records store demand estimates used by organization to plan future blood stock and manage supply more effectively.

3.4 Data Flow Diagram

A data flow diagram is a way of representing a flow of data through a process or a system. The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagrams has no control flow, there are no decision rules and no loops. DFD levels are numbered 0, 1, 2 and occasionally go to even level 3 or beyond. The necessary level of detail depends on the scope of what you are trying to accomplish.

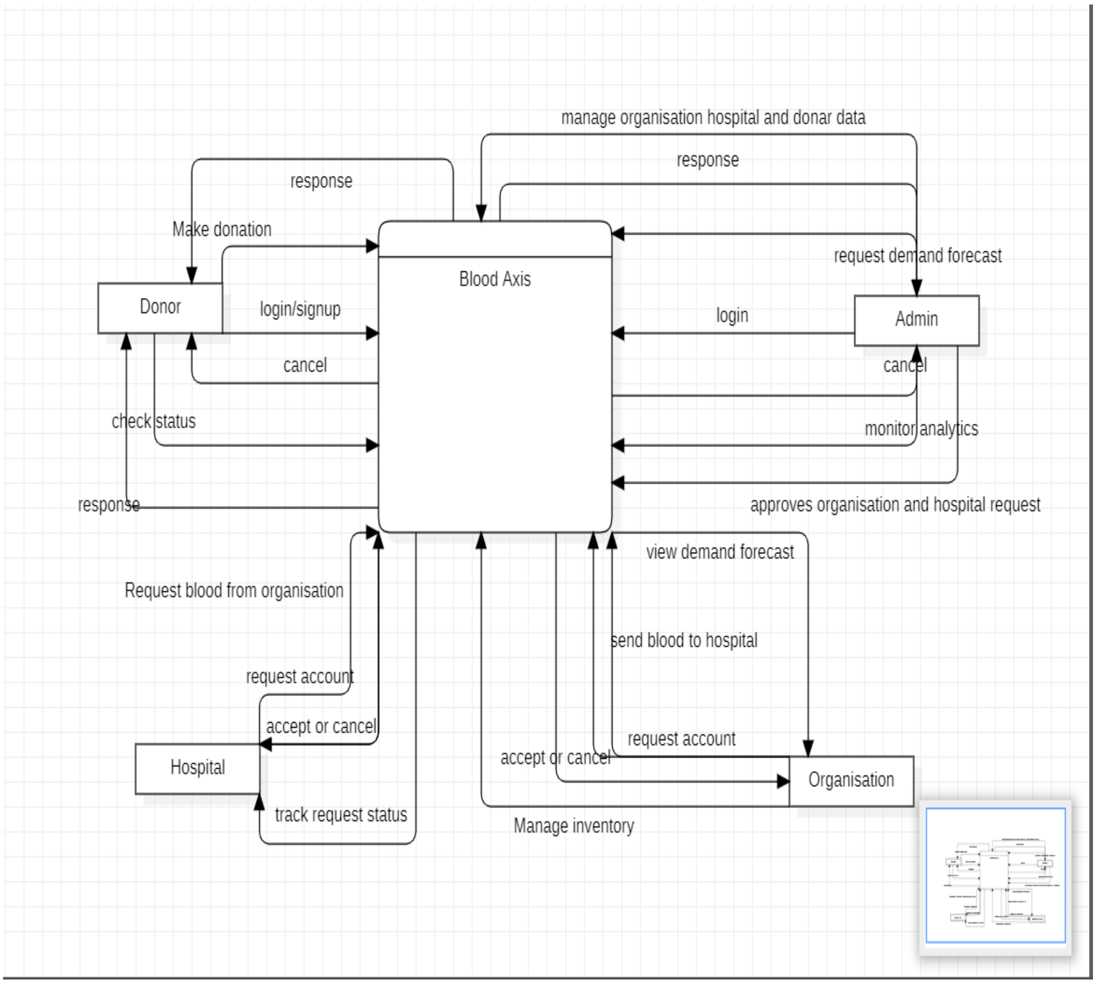

Figure 3.4 1: DFD level 0

The level 0 context diagram shows a simple view of how the Blood Axis system interacts with its main users. The system sits at the center and connects donors, hospitals, organizations, and admin. Donors signup or share their details and submit blood donation information and receive confirmation and status updates. Hospitals send account requests and blood request to the system and receive approvals, rejections and updates on their requests. Organizations send account requests, manage blood stock, through the system and receive approvals inventory updates and demand information. The admin logs in to manage users approve or reject requests, monitor system data, and view demand forecast details. Overall, the diagram shows Blood Axis as a central system that coordinates blood donation, blood requests, inventory management and approvals among all users.

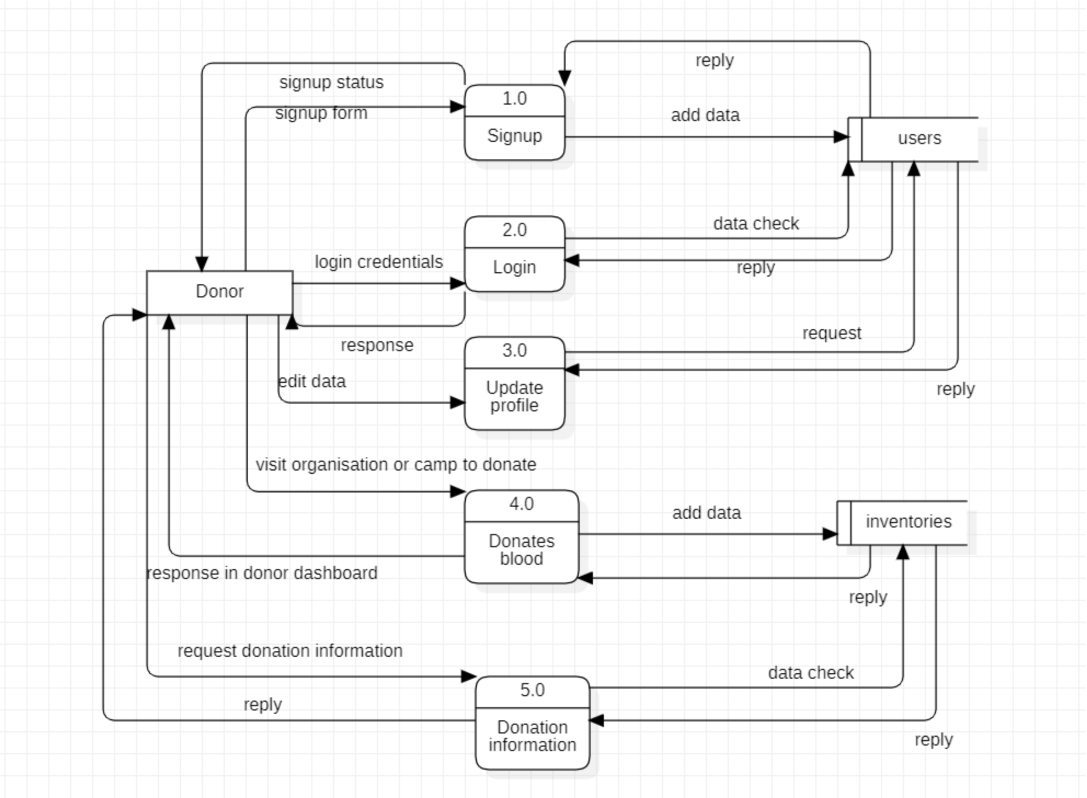

Figure 3.4 2:DFD Level 1 Donor Module

The Donor Module in the Blood Axis System plays a crucial role in the authentication process. It begins when a donor makes a request to either log in or register. Once a user submits their details, the system verifies their information, and if successful, the donor gains access to the platform..

Once the donor is authenticated, they can access various features of the platform, such as donate blood, check donation history, and update profile information.

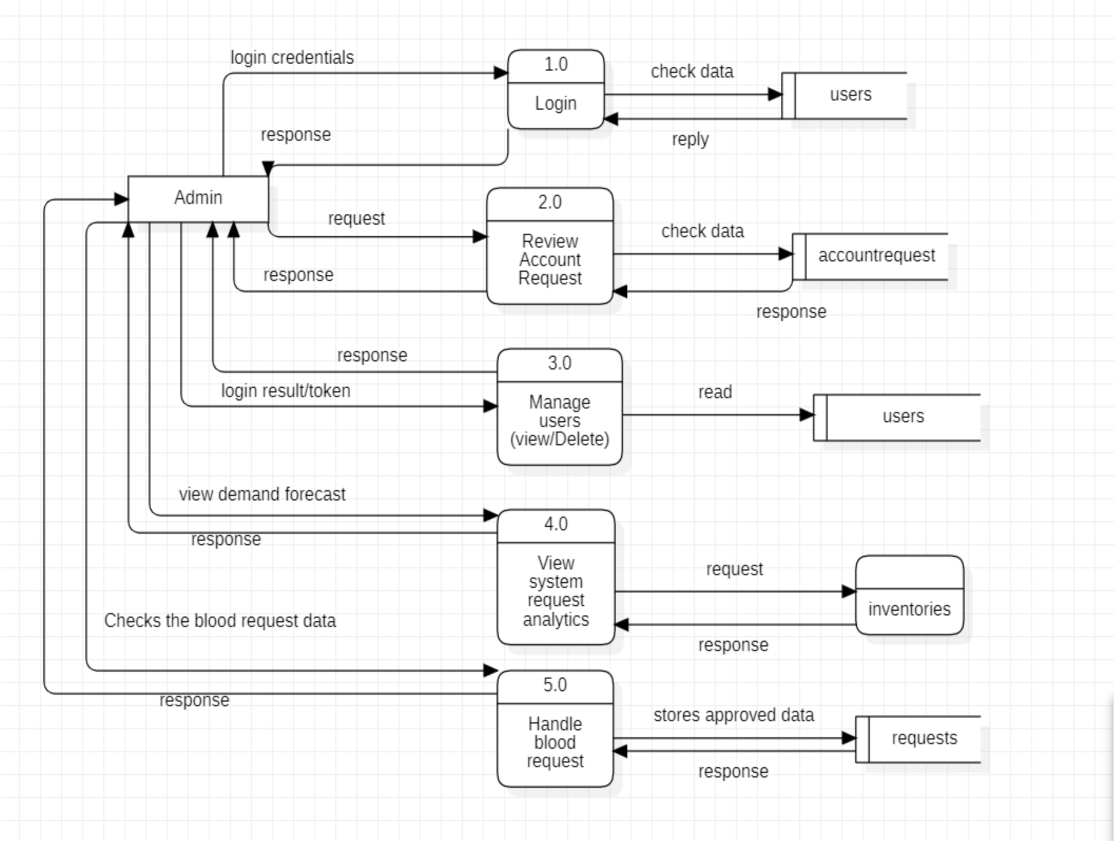

Figure 3.4 3:DFD Level 1 Admin Module

The Admin Module controls and monitors the whole Blood Axis system. It starts when the admin logs in by sending the login details, which are checked against user data and return a login result. After login, the admin reviews account request from hospitals and organizations and approves or rejects them based on stored request data. The admin can also view and remove users by reading user records from the system. To monitor system activity, the admin checks blood request data and inventory records to view system request analytics and demand related information. When needed, the admin handles blood requests by reviewing request details and storing approved decisions in the request records. Through these actions, the admin ensures secure access, correct data handling, and smooth operations of users, requests, and blood inventory across the system.

Chapter 4: Implementation and Discussion

Manoranjan system has two parts: the backend server and also the client-side user view. The server role is to provide authentication and to serve as a gateway to the central database to coordinate between the users.

4.1 Task completed

1. Registration and Login for all roles(donor, organization, hospital, admin) with JWT based authentication
2. Role based access control with protected routes and separate dashboards
3. Donor donation flow with inventory “IN” records linked to organisations
4. Organisations inventory management with “IN/OUT” transactions and stock validations
5. Hospital and organization account request and access to login after admin approval

4.2 Test Case

A test case is a document which has a set of conditions or actions that are performed on software applications in order to verify the expected functionality of the feature. It is well designed and easy understandable steps executed to verify a particular feature of software. Test case describes a specific idea to be tested, without detailing theexact steps to be taken or data to be used. Key purpose of a test case is to ensure of different features within an application are working as expected. Among the different techniques of test cases we have used specification-based technique also known as black-box testing. This type of technique can be used to design test cases in systematic format. These use external features of the software such as technicalspecifications, design, client’s requirements, and more, to derive test case

The prime objective of any software project is to get a high-quality output while reducing the cost and the time required for completing the project. To achieve that, companies test their software before they release it to the market. Documentation plays a critical role in achieving effective software testing.

Test-case Name: Login Functionality

Test-case id	Test-description	Test steps	Expected results	Actual result	Comment
LGN-01	Verify login functionality with valid email, password and role	1. Navigate to login page 2. Enter registered email 3. Enter correct password 4. Select correct role 5. Click Login	User is authenticated, JWT Token is issued, and user is redirected to the correct dashboard for that role	As expected	Pass
		Enter valid username	Credential can be entered	As expected	Pass
		Enter valid password	Credential can be entered	As expected	Pass
		Click on login button	User logged in	As expected	Pass

LGN-02	Verify login functionality with invalid username and password	1. Navigate to login page 2. Enter unregistered or wrong email or password	Login fails, user stays on login page, error message pops up, no token is created	As expected	Pass

		Enter valid username	Credential can be entered	As expected	Pass
		Enter valid password	Credential can be entered	As expected	Pass
		Click on login button	User loggedin	As expected	Pass

Table 1:Test-case Login Functionality

Test-case Name: Signup Functionality

Test case id	Test case description	Test steps	Expected results	Actual results	Comment
REG-01	Successful donor registration	1. Open register as donor page	Donor record is created in users collection with role='donar'	Success message pops out and donar record is stored	Pass

		2. Enter valid details 3. Click Register	success message shown		
	Verify required fields validation	Leave required fields empty. Attempt to register	Registration should fail, and an error message should appear for missing fields	Registration failed and error message appeared	Validation failed for no characters in the input fields

	Verify successful registration	Fill in the require d details with valid data Click the "Register" button	Users should be registered successfully and a confirmation message should be shown	Registration passed and a confirmation message should appear	Pass
	Verify invalid email format validation	Enter an email address without "@" symbol. Attempt to register	Registration should fail, an error message should indicate invalid email format	Registration failed and invalid email format message appeared	Pass

	Verify password length validation	Enter a password with fewer than 4 characters Attempt to register	Registration should fail, and an error message should indicate insufficient password length	Registration failed and message displaying "password too short"	Pass
	Verify successful registration with valid password	Enter a password with length greater than 4 characters.	Users should be registered successfully and confirmation	User registered and confirmation message	Passed as user was registered

		Fill in all the required fields	message should be displayed	appeared	

Table 2: Test-case Signup Functionality

Test-case Name: Donor Donation &Dashboard

Test case id	Test case description	Test steps	Expected results	Actual results	Comment
DON-01	Record blood donation	1. Login as donor 2. Open make donation form 3. Select organisation, blood group and valid quantity 4. Submit form	New "in" inventory record is created linking donor and organization confirmation message is shown	As expected	Pass

Table 3:Test-case Donor Donation and Dashboard

Test-case Name: Organisation Inventory Management

Test case id	Test case description	Test steps	Expected results	Actual results	Comment
ORG-01	Record blood in from donor	Login as organis ation, open record blood in page, select donor, blood group, quantity , save	Inventory "IN" row stored, stock for that blood group increases	As expected	Pass
ORG-02	Prevent OUT when stock is insuffici ent	Login as organis ation and try to record out to hospital with quantity greater than availabl	Inventory is not saved, error message like "Only X ml available" shown	Inventory is not saved	Pass

		e stock			

Table 4:Test-case Add Study Materials

Test-case Name: Admin functionality

Test case id	Test case description	Test steps	Expected results	Actual results	Comment
ADM-01	Approve account request	Login as admin. Open "Account Requests". Select an request and then click approve	Status in account requests changes to "approved", new user created in users	As expected	Passed.
ADM-02	Verify admin can delete user	Login as admin and open user list, select a donor/organis ation/hospital and choose delete	User is removed or marked inactive can no longer login	As expected	Passed
ADM-03	Verify system analytic s	Login as admin and open analytics dashboard.	Admin can see total inventory per blood group and list all blood request	As expected	Passed

Table 5:Test-case Admin functionality

Chapter 5: Conclusion

In conclusion, Manoranjan aims to make learning enjoyable by blending education with entertainment. By implementing features like upload, creating group discussion by video call, the ability to upload study documents, access motivational videos, and more, we are creating a platform that supports students in both their academic and personal growth. Our goal is to foster an engaging and interactive environment where students can learn, share, and connect in a fun and productive way.

5.1 Task Remaining

1. Add a notification module to remind donors when they become eligible again and alert hospitals about low stock blood groups
2. Enhance the hospital and organization dashboard with richer analytics on blood request and transfusion trends to support planning
3. Integrate AI forecasting component directly to UI so admins and organizations can compare forecasted demand with current stock and plan donation camps
4. Enhance the donor dashboard and implement searching nearby blood banks through location access

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